

Practice 12

Deadline: 2 weeks from now. Should be checked onsite (during labs).

Task 1

Download `StandardTests.java` from our course website. This test showcases JUnit test lifecycle methods annotated with `@BeforeAll`, `@BeforeEach`, `@AfterAll` and `@AfterEach`.

1. Run this test and observe the results.
2. Add the class annotation `@TestInstance(TestInstance.Lifecycle.PER_CLASS)`, rerun the tests and observe the results.
3. Explain the difference of results.

Task 2

Based on the Spring Boot application you built in the previous lab:

1. Add an `age` field to the `Student` class.
2. Update `StudentRepository`, `StudentService`, and `StudentRestController` to support the following REST endpoint. Note that, you should leverage `JpaRepository` to define necessary CRUD operations instead of implementing them manually. Check our lecture notes on `Repository` if you need help.
 - `GET /api/students` to get all students.
 - `GET /api/students?email={email}` to get all students whose email contains the given substring.
 - `GET /api/students?age={age}` to get all students less than or equal to the given age.
 - `GET /api/students?email={email}&age={age}` to get all students whose email contains the given substring and age is less than or equal to the given age.
3. Create `StudentRestControllerTest` that uses `MockMvc` to test the above functionalities. Explain your testing logic (e.g., you may want to add more students with diverse attributes for a more robust testing) and show us the test results. Check our lecture notes on `Spring Boot Testing` if you need help.

Run: `StudentRestControllerTest` ×

✓ `StudentRestControllerTest (com.exam)` 658 ms

- ✓ `getStudentsByEmail()` 606 ms
- ✓ `getStudentsByAge()` 21 ms
- ✓ `getStudentsByEmailAndAge()` 16 ms
- ✓ `getAllStudents()` 15 ms

Evaluation

The practice will be checked by teachers or SAs. What will be tested:

1. That you understand every line of your own code, not just copy from somewhere
2. That your program compiles correctly (javac)
3. Correctness of the program logic
4. That the result is obtained in a reasonable time

Late submissions after the deadline will incur a 20% penalty, meaning that you can only get 80% of this practice's score.